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Pymble Ladies' College

Year 12

Extension I Mathematics Trial

11th August 2003

Time allowed : 2 hours plus 5 minutes reading time

Marking guidelines : The marks for each part are indicated beside the question

**Instructions :**

- All questions should be attempted
- All necessary working must be shown
- Start each question on a new page
- Put your name and your teacher's name on each page
- Marks may be deducted for careless or untidy work
- Only approved calculators may be used
- All questions are of equal value
- Diagrams are not drawn to scale
- A standard integral sheet is attached
- DO NOT staple different questions together
- All rough working paper must be attached to the end of the last question
- Staple a coloured sheet of paper to the back of each question
- Hand in this question paper with your answers
- There are seven (7) questions and eight (8) pages in this paper

2

**Question 1**

- |    |                                                                                                                                                               |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| a) | If P is the point ( -3, 5 ) and Q is the point ( 1, -2 ), find the coordinates of the point R which divides the interval PQ externally in the ratio of 3 : 2. | 2 |
| b) | When $(x+3)(x-2)+2$ is divided by $x-k$ , the remainder is $k^2$ . Find the value of $k$ .                                                                    | 2 |
| c) | Solve $\frac{x}{x-3} \geq 1$ .                                                                                                                                | 3 |
| d) | Find the general solution of $\sin \theta = \cos \theta$ .                                                                                                    | 2 |
| e) | Find the exact value of $\int_0^{\frac{\pi}{4}} 2 \sin^2 x \, dx$ .                                                                                           | 3 |

**Question 2** (Start a new page)

a) i) Show that  $x^2 + 4x + 13 = (x + 2)^2 + 9$ .

1

ii) Hence find  $\int \frac{1}{x^2 + 4x + 13} dx$ .

2

b) A stone is projected from the ground with a velocity of  $20 \text{ ms}^{-1}$  at an angle of  $30^\circ$ . Assume that  $\hat{i} = 0$  and  $\hat{j} = -10$ .

i) Prove that :

(1)  $x = 10\sqrt{3}t$

2

(2)  $y = -5t^2 + 10t$

2

ii) Hence find the :

(1) time of flight

1

(2) horizontal range

1

(3) greatest height reached

1

(4) velocity of the particle after  $1\frac{1}{2}$  seconds

2

**Question 3** (Start a new page)

a) Evaluate  $\int_0^{\sqrt{3}} x\sqrt{x^2+1} dx$  using the substitution that  $u = x^2 + 1$ .

3

b) i) Express  $\cos\theta + \sqrt{3}\sin\theta$  in the form  $r\cos(\theta - \alpha)$  where  $r > 0$  and  $0 < \alpha < \frac{\pi}{2}$ .

2

ii) Hence solve  $\cos\theta + \sqrt{3}\sin\theta = 1$  for  $-2\pi \leq \theta \leq 2\pi$ .

2

c) Given  $f(x) = \frac{x-1}{x+2}$ .

i) Write an expression for the inverse function  $f^{-1}(x)$ .

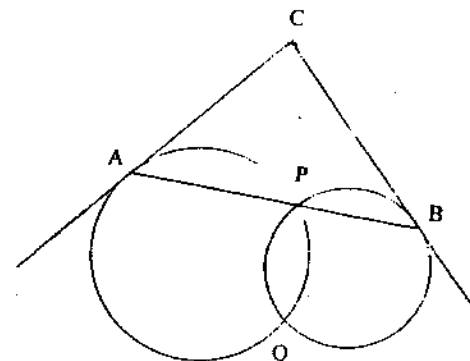
1

ii) Write down the domain and range of  $f^{-1}(x)$ .

1

d) Two circles meet at P and Q. A line APB is drawn through P and the tangents at A and B meet at C. Prove that ACBQ is a cyclic quadrilateral.

3

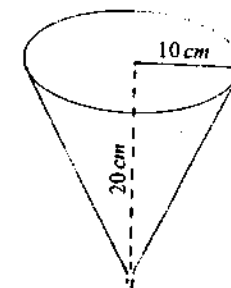


### Question 4 (Start a new page)

- a) Assume that the rate at which a body warms in air is proportional to the difference between its temperature  $T$  and the constant temperature  $A$  of the surrounding air. This rate can be expressed by the differential equation  $\frac{dT}{dt} = -k(T - A)$  where  $t$  is the time in minutes and  $k$  is a constant.
- Show that  $T = A - Ce^{-kt}$  is a solution of the differential equation where  $C$  is a constant. 1
  - A body warms from  $3^\circ\text{C}$  to  $10^\circ\text{C}$  in 15 minutes. The air temperature around the body is  $30^\circ\text{C}$ . Find the temperature of this body after a further 15 minutes have elapsed. Answer correct to the nearest  $^\circ\text{C}$ . 4
  - With the aid of the graph of  $T$  against  $t$ , explain the behaviour of  $T$  as  $t$  becomes large. 1
- b) The acceleration of a particle moving in a straight line is given by  $\ddot{x} = -4x + 8$  where  $x$  is the displacement, in metres, from the origin  $O$  and  $t$  is the time in seconds.
- Show that the particle is moving in simple harmonic motion. 1
  - Write down the centre of motion. 1
  - Show that  $v^2 = 20 + 16x - 4x^2$  given, that the particle is initially at rest at  $x = 5$ . 2
  - Write down the amplitude of the motion. 1
  - Find the maximum speed of the particle. 1

### Question 5 (Start a new page)

- a) Consider the curve  $f(x) = \ln(x+1)$ . Find the gradient(s) of the possible tangent(s) to  $f(x)$  which makes an angle of  $45^\circ$  with the tangent to  $f(x)$  at the point where  $x=1$ . 3
- b) i) Use the table of standard integrals given to find  $\frac{d}{dx} \left[ \ln(x + \sqrt{x^2 + 9}) \right]$ . 1
- ii) Hence use Newton's method to find a second approximation to the root of  $x = \ln(x + \sqrt{x^2 + 9})$ . Take the first approximation as  $x = -4.5$ . 2
- c) Water is running out of a filled conical funnel at the rate of  $5\text{ cm}^3\text{ s}^{-1}$ . The radius of the funnel is  $10\text{ cm}$  and the height is  $20\text{ cm}$ .
- How fast is the water level dropping when the water is  $10\text{ cm}$  deep? 4
  - How long does it take for the water to drop to  $10\text{ cm}$  deep? 2



**Question 6** (Start a new page)

- a) Given  $\theta$  is acute.
- Write  $\sin \frac{\theta}{2}$  in terms of  $\cos \theta$ . 1
  - Prove that  $\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$ . 2
  - If  $\sin \theta = \frac{4}{5}$ , find the value of  $\tan \frac{\theta}{2}$ . 2
- b) Find  $\frac{d}{dx} \cos^{-1}(\sin x)$ . 3
- c) Suppose the roots of the equation  $x^3 + px^2 + qx + r = 0$  are real. Show that the roots are in a geometric progression if  $q^3 = p^3 r$ . 4  
 Hint : let the roots be  $\frac{a}{b}$ ,  $a$  and  $ab$ .

**Question 7** (Start a new page)

- a)i) Prove by mathematical induction that 4
- $$\frac{12}{1 \cdot 3 \cdot 4} + \frac{18}{2 \cdot 4 \cdot 5} + \frac{24}{3 \cdot 5 \cdot 6} + \dots + \frac{6(n+1)}{n(n+2)(n+3)} = \frac{17}{6} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{4}{n+3}$$
- ii) Hence find  $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{6(r+1)}{r(r+2)(r+3)}$ . 1
- b) Consider the variable point  $P(x, y)$  on the parabola  $x^2 = 2y$ .  
 The  $x$  value of  $P$  is given by  $x = t$ ;
- write its  $y$  value in terms of  $t$ . 1
  - write an expression, in terms of  $t$ , for the square of the distance,  $m$ , from  $P$  to the point  $(6, 0)$ . 1
  - hence find the coordinates of  $P$  such that  $P$  is the closest to the point  $(6, 0)$ . 5

\*\*\* End of Paper \*\*\*

### Question 1

a)  $P(-3, 5)$   $Q(1, -2)$   $-3 = 2$   
 $x = \frac{-6-3}{-3+2} = 9$   
 $y = \frac{10+6}{-3+2} = -16$   
 $\therefore R(9, -16)$

b)  $P(k) = (k+3)(k-2)+2 = k^2$   
 $k^2 + k - 6 + 2 = k^2$   
 $k - 4 = 0$   
 $\therefore k = 4$

c)  $\frac{x}{x-3} \geq 1$  ;  $x \neq 3$   
 $x(x-3) \geq (x-3)^2$   
 $x^2 - 3x \geq x^2 - 6x + 9$   
 $3x \geq 9$   
 $x \geq 3$   
 However  $x \neq 3$ ,  $\therefore x > 3$

d)  $\sin \theta = \cos \theta$   
 $\tan \theta = 1$   
 $\theta = \tan^{-1} 1 + n\pi$   
 $\theta = \frac{\pi}{4} + n\pi$

e)  $\int_0^{\frac{\pi}{2}} 2 \sin^2 x \, dx$   
 $= \int_0^{\frac{\pi}{2}} (1 - \cos 2x) \, dx$   
 $= \left[ x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$   
 $= \frac{\pi}{2} - \frac{1}{2} \sin \pi - 0$   
 $= \frac{\pi}{2} - \frac{1}{2} \left( \frac{1}{\sqrt{2}} \right)$   
 $= \frac{\pi}{2} - \frac{\sqrt{2}}{4}$

### Question 2

a) i)  $RHS = (x+2)^2 + 9$   
 $= x^2 + 4x + 4 + 9$   
 $= x^2 + 4x + 13$   
 $= LHS$

iii)  $\int \frac{1}{x^2 + 4x + 13} \, dx = \int \frac{1}{(x+2)^2 + 9} \, dx$   
 $= \frac{1}{3} \tan^{-1} \left( \frac{x+2}{3} \right) + C$

b) i) (1)  $\ddot{x} = 0$   
 $\dot{x} = C_1$   
 When  $t=0$ ,  $\dot{x} = 20 \cos 30^\circ$ ;  $C_1 = 20 \left( \frac{\sqrt{3}}{2} \right) = 10\sqrt{3}$   
 $\Rightarrow \dot{x} = 10\sqrt{3}$   
 $x = 10\sqrt{3}t + C_2$   
 When  $t=0$ ,  $x=0$ ;  $C_2 = 0$   
 $\Rightarrow x = 10\sqrt{3}t$

(2)  $\ddot{y} = -10$   
 $\dot{y} = -10t + C_3$   
 When  $t=0$ ,  $\dot{y} = 20 \sin 30^\circ$ ;  $C_3 = 20 \left( \frac{1}{2} \right) = 10$   
 $\Rightarrow \dot{y} = -10t + 10$   
 $y = -5t^2 + 10t + C_4$   
 When  $t=0$ ,  $y=0$ ;  $C_4 = 0$   
 $\Rightarrow y = -5t^2 + 10t$

iii) (1) When  $y=0$ ;  $-5t^2 + 10t = 0$   
 $-5t(t-2) = 0$   
 $t=0$  or  $t=2$

$\therefore$  Time of flight = 2s

(2) When  $t=2$ ,  $x = 10\sqrt{3}(2) = 20\sqrt{3}$   
 $\therefore$  Horizontal range =  $20\sqrt{3}$  m.

(3) When  $t=1$ ,  $y = -5(1)^2 + 10(1) = 5$   
 $\therefore$  Greatest height = 5m.

(4) When  $t = 1\frac{1}{2}$ ,

$$\dot{x} = 10\sqrt{3} \frac{1}{2}, \quad \dot{y} = -10(1\frac{1}{2}) + 10 = -5 \frac{1}{2}$$

$$\begin{aligned} \Rightarrow v &= \sqrt{(10\sqrt{3})^2 + (-5)^2} \frac{1}{2} \\ &= \sqrt{100 \cdot 3 + 25} \frac{1}{2} \\ &= \sqrt{325} \frac{1}{2} \\ &= 5\sqrt{13} \text{ ms}^{-1} \text{ (going down)}. \quad (2) \end{aligned}$$

### Question 3

$$\begin{aligned} a) \int_0^{\sqrt{3}} x \sqrt{x^2+1} \, dx & \\ &= \int_1^4 \frac{1}{2} \sqrt{u} \, du \frac{1}{2} \\ &= \left[ \frac{1}{3} u^{\frac{3}{2}} \right]_1^4 \frac{1}{2} \\ &= \frac{1}{3} \left[ 4^{\frac{3}{2}} - 1 \right] \frac{1}{2} \\ &= \frac{7}{3} \frac{1}{2} \end{aligned}$$

$$\begin{aligned} u &= x^2 + 1 \\ du &= 2x \, dx \frac{1}{2} \\ x = \sqrt{3}, u &= 3 + 1 = 4 \frac{1}{2} \\ x = 0, u &= 0 + 1 = 1 \frac{1}{2} \end{aligned}$$

(3)

$$\begin{aligned} b) \text{ i) } \cos \theta + \sqrt{3} \sin \theta &= r \cos(\theta - \alpha) \\ \cos \theta + \sqrt{3} \sin \theta &= r \cos \theta \cos \alpha + r \sin \theta \sin \alpha \\ \begin{cases} r \cos \alpha = 1 \\ r \sin \alpha = \sqrt{3} \end{cases} &\Rightarrow \tan \alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3} \quad 1 \\ \begin{cases} r^2 \cos^2 \alpha = 1 \\ r^2 \sin^2 \alpha = 3 \end{cases} &\Rightarrow r^2 = 4 \Rightarrow r = 2 \quad 1 \end{aligned}$$

$$\therefore \cos \theta + \sqrt{3} \sin \theta = 2 \cos(\theta - \frac{\pi}{3}) \quad (2)$$

$$\begin{aligned} \text{ii) } \cos \theta + \sqrt{3} \sin \theta &= 1, \quad -2\pi \leq \theta \leq 2\pi \\ 2 \cos(\theta - \frac{\pi}{3}) &= 1, \quad -\frac{7\pi}{3} \leq \theta - \frac{\pi}{3} \leq \frac{5\pi}{3} \frac{1}{2} \\ \cos(\theta - \frac{\pi}{3}) &= \frac{1}{2} \frac{1}{2} \\ \theta - \frac{\pi}{3} &= \frac{\pi}{3}, \frac{5\pi}{3}, -\frac{\pi}{3}, -\frac{5\pi}{3} \text{ OR } -\frac{7\pi}{3} \frac{1}{2} \\ \theta &= \frac{2\pi}{3}, 2\pi, 0, -\frac{4\pi}{3} \text{ OR } -2\pi \frac{1}{2} \\ \therefore \theta &= -2\pi, -\frac{4\pi}{3}, 0, \frac{2\pi}{3} \text{ OR } 2\pi \quad (2) \end{aligned}$$

$$\begin{aligned} c) \text{ i) } f^{-1}(x) &\Rightarrow x = \frac{y-1}{y+2} \\ xy + 2x &= y - 1 \\ xy - y &= -2x - 1 \frac{1}{2} \\ y(1-x) &= \frac{2x+1}{1-x} \\ f^{-1}(x) &= \frac{2x+1}{1-x} \frac{1}{2} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{ii) Domain : all real } x; x &\neq 1 \frac{1}{2} \quad (1) \\ \text{Range : all real } y; y &\neq -2 \frac{1}{2} \end{aligned}$$

dy Prove that ACBQ is a cyclic quad;

i.e.  $\angle ACB + \angle AQB = 180^\circ$

$$\angle CAB = \angle AQP = \theta \quad (\angle \text{ in alt. segment})$$

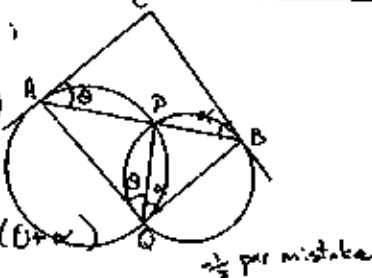
$$\angle CBA = \angle BQP = \alpha \quad (\angle \text{ in alt. segment})$$

$$\angle ACB = 180^\circ - \theta - \alpha \quad (\angle \text{ sum of } \triangle ABC)$$

$$\angle ACB + \angle AQB = (180^\circ - \theta - \alpha) + (\theta + \alpha)$$

$$= 180^\circ$$

$\therefore$  ACBQ is a cyclic quadrilateral.



#### Question 4

i)  $T = A - Ce^{-kt} \Rightarrow T - A = -Ce^{-kt}$   
 $\frac{dT}{dt} = -k(-Ce^{-kt})$   
 $= -k(T - A)$  ①

ii)  $T = 30 - Ce^{-kt}$

When  $t = 0$ ,  $T = 3$ ;

$$3 = 30 - Ce^0$$

$$C = 27$$

$$T = 30 - 27e^{-kt}$$

When  $t = 15$ ,  $C = 10$ ;

$$10 = 30 - 27e^{-k \cdot 15}$$

$$27e^{-15k} = 20$$

$$e^{-15k} = \frac{20}{27}$$

$$-15k = \ln\left(\frac{20}{27}\right)$$

$$k = -\frac{1}{15} \ln\left(\frac{20}{27}\right)$$

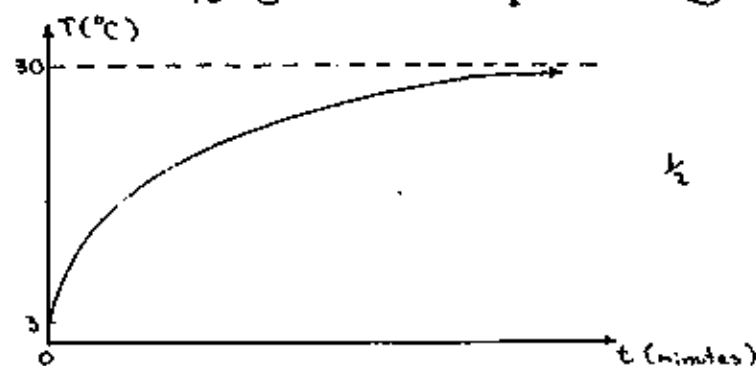
When  $t = 30$ ;

$$T = 30 - 27e^{-30k}$$

$$= 15.18 \dots$$

$$= 15^\circ \text{C}$$

iii)



As  $t$  becomes large,  $T$  approaches  $30^\circ \text{C}$ .

- b) i)  $\ddot{x} = -4x + 8$   
 $\ddot{x} = -4(x - 2)$  ①  
 $\Rightarrow \ddot{x} = -\omega^2 x \Rightarrow \text{S.H.M}$
- ii) Centre of motion = 2 ①
- iii)  $\ddot{x} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = -4x + 8$   
 $\frac{1}{2} v^2 = -2x^2 + 8x + C$  ②  
 When  $x = 5, v = 0$ ;  
 $0 = -2(25) + 40 + C$  ①  
 $C = 10$  ②  
 $\frac{1}{2} v^2 = -2x^2 + 8x + 10$  ②  
 $v^2 = -4x^2 + 16x + 20$   
 $v^2 = 20 + 16x - 4x^2$  ②
- iv) Amplitude =  $5 - 2 = 3 \text{ m}$  ①
- v) Max. velocity when  $x = 2$ ;  
 $v^2 = 20 + 16(2) - 4(4)$  ②  
 $v^2 = 36$   
 $v = \pm 6$   
 $\therefore \text{max. speed} = 6 \text{ ms}^{-1}$  ①

### Question 5

- a)  $f(x) = \ln(x+1)$   
 $f'(x) = \frac{1}{x+1}$  ②  
 $f'(1) = \frac{1}{2}$  ②
- 1/2 if only find value of m.  
 $\tan 45^\circ = 1 = \left| \frac{\frac{1}{2} - m}{1 + \frac{1}{2}m} \right|$  ②
- $1 + \frac{1}{2}m = \frac{1}{2} - m$  OR  $-1 - \frac{1}{2}m = \frac{1}{2} - m$   
 $\frac{3}{2}m = -\frac{1}{2}$  ②  $\frac{1}{2}m = \frac{3}{2}$   
 $m = -\frac{1}{3}$  ②  $m = 3$  ②
- For all  $x, x > -1, f'(x) > 0$ , ②  
 $\therefore m = 3$  ②
- b) i)  $\frac{d}{dx} [\ln(x + \sqrt{x^2 + 9})]$   
 $= \frac{1}{\sqrt{x^2 + 9}}$  ①
- ii)  $x - \ln(x + \sqrt{x^2 + 9}) = 0$  ②  
 $x_1 = 4.5 \quad \frac{(-4.5) - \ln(-4.5 + \sqrt{(-4.5)^2 + 9})}{1 - \frac{1}{\sqrt{(-4.5)^2 + 9}}}$   
 $= 0.9028$   
 $= 0.90$  ②
- c) i)  $V = \frac{1}{3} \pi r^2 h$  ②  $r = \frac{1}{2} h$  ②  
 $= \frac{1}{3} \pi \left( \frac{h}{2} \right)^2 h$   
 $= \frac{1}{12} \pi h^3$  ②  
 $\frac{dV}{dh} = \frac{1}{4} \pi h^2$  ②  
 $\frac{dh}{dt} = \frac{dV}{dV} \times \frac{dV}{dt}$  ②  
 $= \frac{4}{\pi h^2} \times (-5)$  ②  
 $= \frac{-20}{\pi h^2}$  ②  
 When  $h = 10, \frac{dh}{dt} = \frac{-1}{5\pi} \text{ cm/s}$  ④

$$ii) \frac{dv}{dt} = -5$$

$$v = -5t + C$$

$$\text{When } t = 0, v = \frac{1}{12} \pi (20)^3 = \frac{2000\pi}{3}$$

$$\frac{2000\pi}{3} = 0 + C$$

$$\Rightarrow v = -5t + \frac{2000\pi}{3}$$

$$\text{When } h = 10, v = \frac{1}{12} \pi (10)^3 = \frac{250\pi}{3}$$

$$\frac{250\pi}{3} = -5t + \frac{2000\pi}{3}$$

$$5t = \frac{1750\pi}{3}$$

$$t = \frac{350\pi}{3}$$

$$= 367 \text{ s (nearest whole number)} \quad (2)$$

### Question 6

$$a) i) \cos \theta = 1 - 2 \sin^2 \frac{\theta}{2}$$

$$2 \sin^2 \frac{\theta}{2} = 1 - \cos \theta$$

$$\sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2}$$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\sin \frac{\theta}{2} = \sqrt{\frac{1 - \cos \theta}{2}} \quad \frac{1}{2} \theta \text{ is acute} \quad (1)$$

$$ii) \text{ Prove that } \tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\text{RHS} = \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{1 + (2 \cos^2 \frac{\theta}{2} - 1)}$$

$$= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}}$$

$$= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}$$

$$= \tan \frac{\theta}{2}$$

$$= \text{LHS}$$

$$\text{LHS} = \tan \frac{\theta}{2}$$

$$= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}$$

$$= \sqrt{\frac{1 - \cos \theta}{2}} \times \frac{1}{\sqrt{\frac{1 + \cos \theta}{2}}}$$

$$= \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \frac{1 + \cos \theta}{1 + \cos \theta}$$

$$= \sqrt{\frac{1 - \cos^2 \theta}{(1 + \cos \theta)^2}}$$

$$= \sqrt{\frac{\sin^2 \theta}{(1 + \cos \theta)^2}}$$

$$= \frac{\sin \theta}{1 + \cos \theta}$$

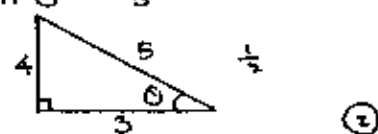
$$= \text{RHS} \quad (2)$$

$$iii) \tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{\frac{4}{5}}{1 + \frac{3}{5}}$$

$$= \frac{1}{2}$$

$$\sin \theta = \frac{4}{5}$$



$$b) \frac{d}{dx} \cos^{-1}(\sin x)$$

$$= \frac{-\frac{d}{dx} \sin x}{\sqrt{1 - (\sin x)^2}}$$

$$= \frac{-\cos x}{\sqrt{\cos^2 x}}$$

$$= \frac{-\cos x}{|\cos x|}$$

$$= \begin{cases} -1 & \text{for } x \text{ in 1st \& 4th quadrant} \\ 1 & \text{for } x \text{ in 2nd \& 3rd quadrant} \end{cases}; \cos x \neq 0$$

$$(3)$$

$$\begin{aligned}
 & x^3 + px^2 + qx + r = 0 \\
 & \text{Let the roots be } \frac{a}{b}, a \text{ and } ab, \\
 & \text{then } \left[ \frac{a}{b} + a + ab = -p \right. \quad \text{--- (1)} \quad \frac{1}{2} \\
 & \quad \left[ \left(\frac{a}{b}\right)(a)(b) = -r \Rightarrow a^3 = -r \right. \quad \text{--- (2)} \quad \frac{1}{2} \\
 & \quad \left. \left(\frac{a}{b}\right)(a) + (a)(ab) + \left(\frac{a}{b}\right)(ab) = q \right. \quad \text{--- (3)} \quad \frac{1}{2} \\
 & \quad \Rightarrow \frac{a^2}{b} + a^2b + a^2 = q \\
 & \quad a^2 \left( \frac{1}{b} + b + 1 \right) = q \\
 & \quad q^3 = \left[ a^2 \left( \frac{1}{b} + b + 1 \right) \right]^3 \\
 & \quad q^3 = a^6 \left( \frac{1}{b} + b + 1 \right)^3 \\
 & \text{From (1) \& (2), } p^3 r = (-p)^3 (-r) \\
 & \quad = \left( \frac{a}{b} + a + ab \right)^3 (a^3) \\
 & \quad = \left[ a \left( \frac{1}{b} + 1 + b \right) \right]^3 (a^3) \\
 & \quad = \left[ a^3 \left( \frac{1}{b} + b + 1 \right) \right]^3 (a^3) \\
 & \quad = a^6 \left( \frac{1}{b} + b + 1 \right)^3 \\
 & \quad = q^3 \quad \text{--- (4)}
 \end{aligned}$$

### Question 7

a) i) Prove by induction that the statement

$$\frac{12}{1 \cdot 3 \cdot 4} + \frac{18}{2 \cdot 4 \cdot 5} + \frac{24}{3 \cdot 5 \cdot 6} + \dots + \frac{6(n+1)}{n(n+2)(n+3)} = \frac{17}{6} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{4}{n+3}$$

is true.

Step 1: Prove that the statement is true for  $n=1$ ;

$$\begin{aligned}
 \text{LHS} &= \frac{6(1+1)}{1(1+2)(1+3)} = \frac{12}{1 \cdot 3 \cdot 4} = \frac{1}{1} \\
 \text{RHS} &= \frac{17}{6} - \frac{1}{1+1} - \frac{1}{1+2} - \frac{4}{1+3} \\
 &= \frac{17}{6} - \frac{1}{2} - \frac{1}{3} - 1 \\
 &= 1 \\
 &= \text{LHS}
 \end{aligned}$$

Step 2: Assume the statement is true for  $n=k$ .

Step 3: Prove that the statement is true for  $n=k+1$ ;

$$\text{i.e. prove } \frac{12}{1 \cdot 3 \cdot 4} + \frac{18}{2 \cdot 4 \cdot 5} + \dots + \frac{6(k+2)}{(k+1)(k+3)(k+4)}$$

$$\begin{aligned}
 \text{LHS} &= \frac{12}{1 \cdot 3 \cdot 4} + \frac{18}{2 \cdot 4 \cdot 5} + \frac{24}{3 \cdot 5 \cdot 6} + \dots + \frac{6(k+1)}{k(k+2)(k+3)} + \frac{6(k+2)}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+1} - \frac{1}{k+2} - \frac{4}{k+3} + \frac{6(k+2)}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+2} - \frac{1}{k+3} - \frac{3}{k+3} - \frac{1}{k+1} + \frac{6(k+2)}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+2} - \frac{1}{k+3} + \frac{(-3)(k+1)(k+4) - (k+3)(k+4) - (k+1)}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+2} - \frac{1}{k+3} + \frac{-3k^2 - 15k - 12 - k^2 - 7k - 12 - 6k - 12}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+2} - \frac{1}{k+3} + \frac{-4k^2 - 16k - 12}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+2} - \frac{1}{k+3} + \frac{(-4)(k^2 + 4k + 3)}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+2} - \frac{1}{k+3} + \frac{(-4)(k+1)(k+3)}{(k+1)(k+3)(k+4)} \\
 &= \frac{17}{6} - \frac{1}{k+2} - \frac{1}{k+3} - \frac{4}{k+4} \\
 &= \text{RHS}
 \end{aligned}$$

Step 4: ...

(4)

$$\text{ii) } \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{r(r+2)(r+3)} = \frac{11}{6} \quad \text{①}$$

$$x^2 = 2y$$

$$\text{iv) } x = t; \quad t^2 = 2y \Rightarrow y = \frac{1}{2}t^2 \quad \text{①}$$

$$\begin{aligned} \text{v) } m^2 &= (t-6)^2 + \left(\frac{1}{2}t^2 - 0\right)^2 \\ &= (t-6)^2 + \left(\frac{t^2}{2}\right)^2 \quad \frac{1}{2} \\ &= t^2 - 12t + 36 + \frac{t^4}{4} \quad \frac{1}{2} \quad \text{①} \end{aligned}$$

$$\text{vi) } \frac{dm}{dt} = 2t - 12 + t^3 = 0 \quad - \text{①}$$

$$(t-2)(t^3+2t+6) = 0 \quad - \text{②} \quad \frac{1}{2}$$

$$t-2 = 0$$

$$t = 2 \quad \frac{1}{2}$$

$$\therefore P(2, 2) \quad \frac{1}{2}$$

$$\text{①} \Rightarrow \text{Let } P(t) = t^3 + 2t - 12$$

$$P(2) = 8 + 4 - 12 = 0 \quad \frac{1}{2}$$

$$\begin{array}{r} t^3 + 2t + 6 \\ t-2 \overline{) t^3 + 0 + 2t - 12} \\ \underline{t^3 - 2t^2} \phantom{+ 6} \\ 2t^2 + 2t - 12 \\ \underline{2t^2 - 4t} \phantom{+ 6} \\ 6t - 12 \\ \underline{6t - 12} \\ 0 \end{array}$$

$$\text{②} \Rightarrow t^2 + 2t + 6 = 0 \quad \frac{1}{2}$$

$$\Delta = (2)^2 - 4(1)(6) < 0 \quad \frac{1}{2}$$

$$\text{So } t^2 + 2t + 6 = 0 \text{ has no solutions.} \quad \text{⑤}$$